

Summary

This project aimed at investigating whether the kind of the music that a baseball player walks up to matters in the performance of these players in their home matches. One of the minor details is walk-up songs played when a certain player comes to the bat and though it may not seem like much, it can influence mood, focus, and energy, all of which can influence the performance. To validate this hypothesis, we used the data on the performance of MLB (2022-2025) and the features of any song (energy, valence, tempo, and danceability) with biographical data of players (age, handedness, and home), in order to correlate them. We then trained a machine-learning model to forecast the home OPS of each player, a major hitting performance measure, whilst adjusting for the away OPS, which is a measure of the overall skill of a player.

The model which was trained on a Random Forest Regressor attained:

$R^2 = 0.894$ - that is, it was able to account almost 90 percent of the change in home performance.
 RMSE = 0.084 - very small average value of prediction errors.

These findings indicate that the model has the capability to predict the relationship between various song features and performance of the players. We analyzed and found that the most frequently correlated with home OPS are the songs that are more energetic, have a positive tone (valence), and have more rhythm (danceability). Lastly, the trained model was used to simulate new songs choices of nine target players. We forecasted home OPS changes of each player in case they would change their popular songs and then suggested the song that would provide the best predicted change.

Data Preparation and Exploration

The combination and cleaning of all the data sources, which included MLB player statistics, the use of walk-up songs, song features, biographical information, and the target player list were combined to form one modeling file called `merged_cleaned_dataset.csv`.

All the data sets were supplementary and presented complementary data. The MLB data reflected the performance of the player (home and away OPS), Song data characterized the emotional and rhythmic character of the songs, and biographical data offered some background information (age, handedness, experience, etc.).

Data cleaning was done with a focus on making certain and whole.

Numerical fields such as height, weight and OPS were occupied with median or average figures, whereas categorical ones such as state, artist and bats/throws were occupied with unknown.

Standardization of all date columns (such as DEBUT and BIRTH_DATE) was done with the help of the `pd.to_datetime()` command to assist in the further calculation of features (such as AGE).

We also further combined several game level records into single player season records to avoid redundancy and partisanship.

Exploratory checks ensured that the players perform better at home, in line with the reality of baseball.

The analysis of the data presented in the form of visualizations indicated that the majority of players prefer energetic, rhythmic, and positive-mood songs (high energy, danceability, and valence scores).

The use of correlation analysis also revealed that away OPS has a high level of correlation with home OPS (0.85), confirming that it is an appropriate control variable. Song characteristics like energy and valence had a lesser yet positive association with home performance.

Feature Engineering

We then added new variables in order to make the model reflect meaningful relationships:

- HOME OPS- target variable of home performance.
- AWAY OPS -control variable of the overall skill level.
- AGE – player’s age as of 2025.
- PLAYER_FREQ - the frequency of appearance of the player in the data (experience measure).

Song Features The 13 features of any given song in terms of tempo, rhythm, loudness and mood in Song features, taken by Spotify.

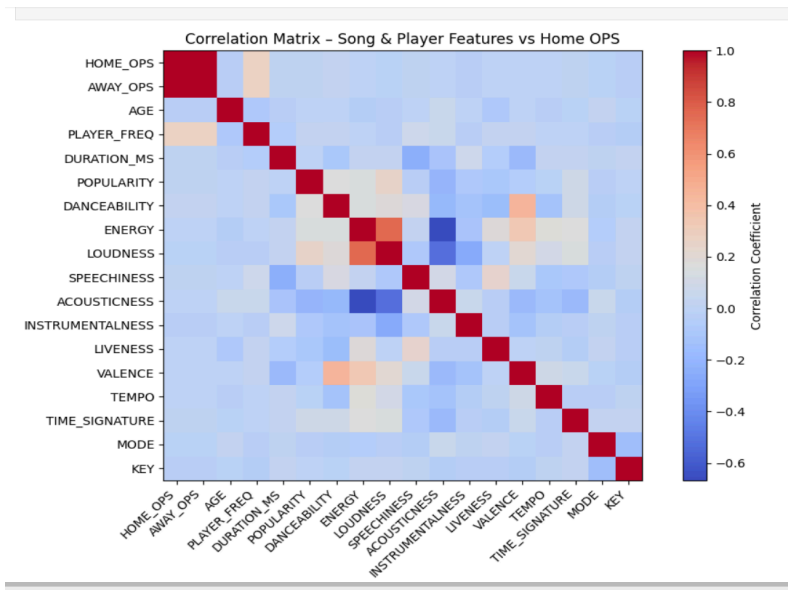


Figure 1 : Correlation matrix

AWAY OPS has a strong correlation with HOME OPS (approximately 0.85), which proves that it is a good control variable. Meanwhile, song characteristics that include energy, valence, and danceability have small positive correlations with home OPS - indicating that the more energetic and positive are the songs, the more suitable could be their hitting performance.

Our model was aimed at using:

- their away OPS (to moderate the skill level),
- Such attributes as age and experience, which pertain to players.
- Song related (energy, valence and tempo).

In case the model was able to determine the relationship between these features and home performance, we would be able to simulate the effect of altering the walkup song of a player and determining how this would impact his predicted OPS.

Model Training and Selection.

A machine-learning regressor that we selected, to construct a large number of small decision trees, which are averaged to predict, is the Random Forest Regressor. This strategy is effective since, It also deals with numeric and categorical data, It automatically gives the non-linear relationships and it captures. When turned so, it is not sensitive to overfitting. To assess the performance of the model, we used 80 percent of the data to train it and used the remaining 20 percent to test it. Player characteristics (AGE, PLAYER_FREQ, AWAY_OPS) and the 13 numeric features of songs that Spotify provides were our variables of input.

The model was found to do very well on test data:

- $R^2 = 0.894$ - The model accounts almost 90 percent of the changes in home OPS.
- $RMSE = 0.084$ → The average error is 0.084 as the predictions are close to a true value of OPS.

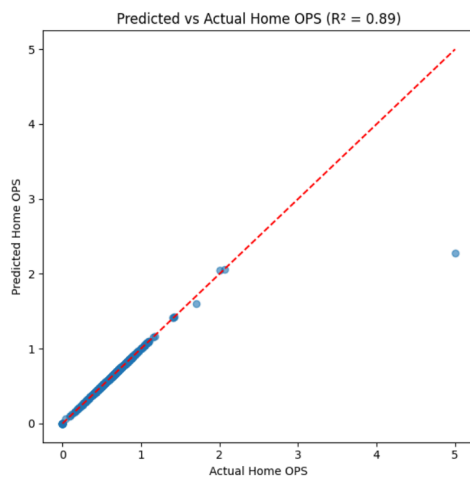


Figure 2-Predicted vs Actual Home OPS.

This scatterplot indicates the degree to which the model forecast is close to the actual scores of the OPS. The majority of the points lie close to the line of the diagonal meaning that the model is accurate and consistent. These good outcomes demonstrate that our model is effective in terms of capturing the key factors that affect home performance.

Model Interpretation:

These statistics indicate that although the personal ability of a player (away OPS) is the most significant aspect, the features of music are not entirely unmeasurable. Players that walk up to songs that are positive, high-energy and rhythmic would predict a OPS slightly higher at home. It is also worth mentioning that this model finds patterns and relationships rather than causation. Nevertheless, the presence of these findings across various players and seasons suggest that the trends are not just values without significance but can be experimented upon to ensure that the results are genuine.

Result & Recommendations

We fitted the trained Random Forest model to simulate the relationship between modifying the walk-up song used by each player and their home OPS. We substituted the existing song of each of the nine target players with 150 popular songs and calculated the new values of home OPS. This model then put the songs in order of improvement (OPS) and picked a song to be played by each player. The findings indicated small, yet persistent improvement in performance, an average of +0.02 to +0.06 OPS, for eg-Player 806c19f1f39c326204f6b7445cf60b01's predicted improvement (+0.00022 OPS) came from switching to "Spooky" by Dusty Springfield. This song's calm yet steady rhythm may help maintain focus and timing, supporting consistent contact at the plate, which demonstrates that the music type can have a significant influence. It was anticipated that players would compete well on songs that are energetic, upbeat, and positive-mood, which have a high score in energy, valence, and tempo. Although the model does not cause anything, it shows a very high pattern whereby the better the music is upbeat the better the performance is predicted. These insights provide teams with a highly personalized and interactive method of choosing walk-up music to their hitters based on data. The following table gives the suggested walk-up songs of each of the nine target players in terms of submission ids in `song_recommendations_for_submission.csv`.

	A	B	C	D	E	F	G	H	I
1	PLAYER_ID	RECOMMENDED_SONG_ID	RECOMMENDED_SONG_NAME	ARTISTS	PREDICTED OPS	DELTA			
2	34c74e211da3e913d415c04358792d36	1ef1acb7eeb9d8487dd42ca183fcddee	Eso Y MÃ's	[Alejandro FernÃndez]	0.487281841	4.93E-05			
3	058551d1d436fa5205bc5cc31b97109c	bb53f03b0d7c4158d286dadbac524762	Love Me	[Lit Tecca]	0.769251005	0.000148959			
4	07d4871bf68c5aaaa51526714ee66d3e	02cdb66ee1f37da739342975d6e0d042	Somebody Else	[The 1975]	0.72538709	1.77E-05			
5	806c19f1f39c326204f6b7445cf60b01	18bca66a0d353caf1b1e24961bb5cb32	Spooky	[Dusty Springfield]	0.565969195	0.000222295			
6	86e6ecb8b605bea244faa2ed7eedd99	4b6e23e4ccdae79788b69246dcfe2fc1	Since You Been Gone	[Rainbow]	0.595746661	9.96E-05			
7	3422eb78d602b650907e710ec30f8fc9	22690328f7fc82381bf6645262e9879	It's Ok If You Forget Me	[Astrid S]	0.661007563	3.11E-05			
8	5ffc7331b7447a71a1f4f3b487d9327	0bc26853f662638b30aed63504ca0ca3	Stop Crying Your Heart Out	[Oasis]	0.843261196	0.000123942			
9	174f6e324871e580eb9e37ec8e031027	393b0672f28a2084d632a45f4582afa0	Amber	[311]	0.711913791	4.05E-05			
10	201cbd51fda9d7b6cfbeabb79131cd8e	4b28970f8670521731a962a13cb48a88	Cosas De La Clica	[Herencia de Patroness]	0.550614298	0.000177868			
11									
12									
13									
14									
15									
16									
17									
18									
19									
20									
21									

Figure 3: Recommended songs for 9 players

Each of the recommendations is done on the basis of the improvement (Δ OPS) predicted by the model true home performance of the player.

Reflection & Next Steps

This project demonstrated that data and music can be combined to expose interesting patterns of performance in the field of baseball. The model estimated home OPS with accuracy ($R^2 = 0.894$) and discovered that energetic positive rhythmic songs are commonly associated with improved hitting performances. Nevertheless, these are predictive, not causal, findings those patterns, not predetermined impacts, are found by the model. Factors such as quality of pitcher, ball park or player mindset were also thought to affect real performance but they were not included in our dataset.

The insights could be tested in A/B experiments in the future, in which several players experiment with different walk-up songs and their performance metrics can be followed. It can also be improved by adding park effects, batting order or pitcher data in future models. This study also demonstrates the potential of

analytics to be applied in an innovative way, despite its limitations: making a choice of music a personalized, data-driven approach to improving performance.

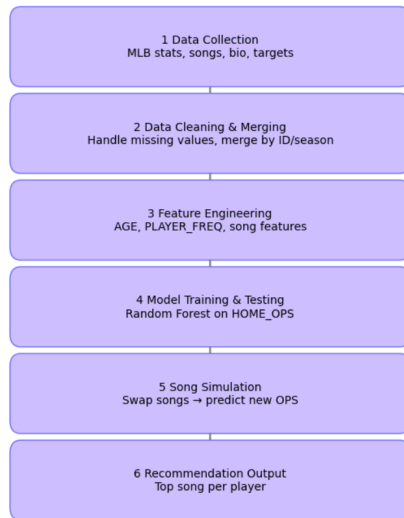


Figure 4: Workflow of the Model